

Position Analysis:

$$\vec{OA} = x_A + iy_A$$

$$x_q = 0$$

$$\rightarrow x_A = L_2 \cos \theta_2$$

$$y_q = L_3$$

$$\rightarrow y_A = L_2 \sin \theta_2$$

Loop O A Q:

$$\vec{OA} = \vec{OQ} + \vec{QA}$$

$$x_A + iy_A = x_q + iy_q + x e^{i\theta_3}$$

$$x e^{i\theta_3} = [x_A - x_q] + i[y_A - y_q]$$

$$\rightarrow x = \sqrt{(x_A - x_q)^2 + (y_A - y_q)^2}$$

$$\rightarrow \cos \theta_3 = \frac{x_A - x_q}{x}, \sin \theta_3 = \frac{y_A - y_q}{x}$$

Loop O B Q:

$$\vec{OB} = \vec{OQ} + \vec{QB}$$

$$x_B + iy_B = x_q + iy_q + L_4 e^{i\theta_3}$$

equating real parts:

$$\rightarrow x_B = x_q + L_4 \cos \theta_3$$

equating imaginary parts:

$$\rightarrow y_B = y_q + L_4 \sin \theta_3$$

Loop O C B:

$$x_c + iy_c = x_B + iy_B + L_5 e^{i\theta_4}$$

equating imaginary parts:

$$y_c = y_B + L_5 \sin \theta_4$$

$$\rightarrow \sin \theta_4 = \frac{y_c - y_B}{L_5}$$

equating real parts:

$$\rightarrow x_c = x_B + L_5 \cos \theta_4$$

Loop O D C:

$$\vec{OD} = \vec{OC} + \vec{CD}$$

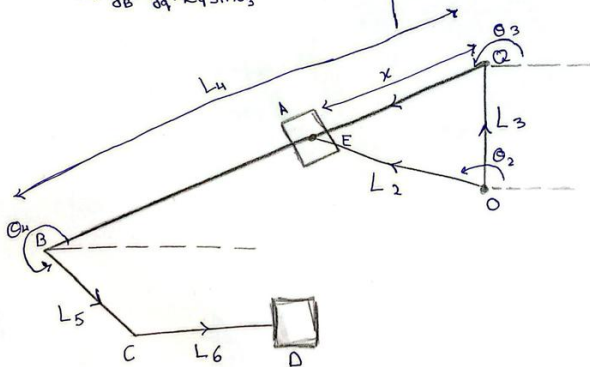
$$x_0 + iy_0 = x_c + iy_c + L_6 e^{i\theta_1}$$

equating imaginary parts:

$$\rightarrow y_0 = y_c + L_6 \sin \theta_1$$

equating real parts:

$$\rightarrow x_0 = x_c + L_6 \cos \theta_1$$



Velocity Analysis:

Point A:

$$V_A = L_2 \omega_2 (e^{i\theta_2}) \quad , \quad \begin{aligned} V_A^x &= L_2 \omega_2 \sin \theta_2 \\ V_A^y &= L_2 \omega_2 \cos \theta_2 \end{aligned}$$

Point E:

$$\vec{V}_E = \vec{V}_A + \vec{V}_{EA} = \kappa \omega_3 (e^{i\theta_3}) = V_A^x + i V_A^y + V_{EA} (e^{i\theta_3}) e^{-i\theta_3}$$

$$-i \kappa \omega_3 = [V_A^x + i V_A^y] e^{-i\theta_3} + V_{EA}$$

$$V_{EA} = -V_A^x \cos \theta_3 - V_A^y \sin \theta_3$$

$$\omega_3 = \frac{V_A^x \sin \theta_3 - V_A^y \cos \theta_3}{\kappa}$$

Point B:

$$\vec{V}_B = \vec{V}_A + \vec{V}_{AB} \quad , \quad V_B = L_4 \omega_3 (e^{i\theta_3})$$

$$V_B^x = -L_4 \omega_3 \sin \theta_3 \quad , \quad V_B^y = L_4 \omega_3 \cos \theta_3$$

Point C:

$$\vec{V}_C = \vec{V}_B + \vec{V}_{CB} \quad , \quad V_C = V_B^x + i V_B^y + (\omega_4 L_5) e^{i\theta_4}$$

$$\text{Real: } V_C = V_B^x - \omega_4 L_5 \sin \theta_4$$

$$\text{Imag: } \omega_4 = \frac{-V_B^y}{L_5 \cos \theta_4}$$

Acceleration Analysis:

Point A:

$$\vec{A}_A = (-\omega_2^2 L_1 + i\alpha_2 r_2) e^{i\theta_2} = A_A^x + iA_A^y$$

$$A_A^x = -L_1 \omega_2^2 \cos \theta_2 - L_2 \alpha_2 \sin \theta_2$$

$$A_A^y = -L_1 \omega_2^2 \sin \theta_2 + L_2 \alpha_2 \cos \theta_2$$

Point E:

$$\vec{u}_E = e^{i\theta_3}, \vec{u}_n = -ie^{i\theta_3}, A_{EA}^n = -2V_{EA}\omega_3$$

$$\vec{A}_E = \vec{A}_A + \vec{A}_{EA}$$

$$\vec{A}_{EA} = A_{EA}^n \vec{u}_n + A_{EA}^{sl} \vec{u}_E$$

$$\vec{A}_E = A_A^x + iA_A^y + \vec{A}_{EA}$$

$$-\omega_3^2 x = A_A^x \cos \theta_3 + A_A^y \sin \theta_3 + A_{EA}^{sl}$$

$$\alpha_3 = \frac{-A_A^x \sin \theta_3 + A_A^y \cos \theta_3 + 2V_{EA}\omega_3}{x}$$

Point B:

$$\vec{A}_B = \vec{A}_O + \vec{A}_{OB}$$

$$\vec{A}_B = (-\omega_3^2 L_4 + i\alpha_3 L_4) e^{i\theta_3}$$

$$A_B^x = -\omega_3^2 L_4 \cos \theta_3 - \alpha_3 L_4 \sin \theta_3$$

$$A_B^y = -\omega_3^2 L_4 \sin \theta_3 + \alpha_3 L_4 \cos \theta_3$$

Point C:

$$A_C = \vec{A}_B + \vec{A}_{CB}$$

$$A_C = A_B^x + iA_B^y + \vec{A}_{CB}$$

$$A_C = A_B^x + iA_B^y + (-\omega_4 L_5 + i\alpha_4 L_5) e^{i\theta_4}$$

$$\text{Real: } A_C = A_B^x - \omega_4 L_5 \cos \theta_4 - \alpha_4 L_5 \sin \theta_4$$

$$\text{Imag: } \alpha_4 = \frac{-A_B^y + \omega_4 L_5 \sin \theta_4}{L_5 \cos \theta_4}$$

Force analysis

$$g_2 = \frac{r_2}{2}$$

$$g_4 = r_4/2$$

$$g_5 = r_5/2$$

$$g_6 = r_6/2$$

$$A_{g2} = A_0 + A_{g0}$$

$$A_{g2} = [-\omega_2^2 g_2 + i \alpha_2 g_2] e^{i\theta_2}$$

$$A_{g2}^x = -\omega_2^2 g_2 \cos \theta_2 - \alpha_2 g_2 \sin \theta_2$$

$$A_{g2}^y = -\omega_2^2 g_2 \sin \theta_2 + \alpha_2 g_2 \cos \theta_2$$

$$A_{g4} = A_0 + A_{g0} = [-\omega_3^2 g_4 + i \alpha_3 g_4] e^{i\theta_3}$$

$$A_{g4}^x = -\omega_3^2 g_4 \cos \theta_3 - \alpha_3 g_4 \sin \theta_3$$

$$A_{g4}^y = -\omega_3^2 g_4 \sin \theta_3 + \alpha_3 g_4 \cos \theta_3$$

$$A_{g5} = A_{B^x} + A_{B^y} J + [-\omega_4^2 g_5 + i \alpha_4 g_5] e^{i\theta_4}$$

$$A_{g5}^x = A_{B^x} - \omega_4^2 g_5 \cos \theta_4 - \alpha_4 g_5 \sin \theta_4$$

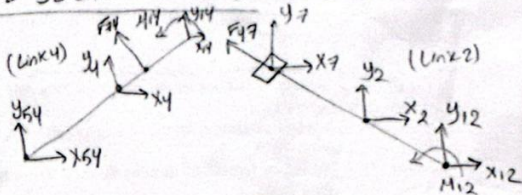
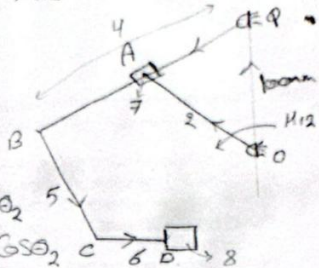
$$A_{g5}^y = A_{B^y} - \omega_4^2 g_5 \sin \theta_4 + \alpha_4 g_5 \cos \theta_4$$

$$A_{g6} = A_{C^x} + A_{C^y} J + [-\omega_6^2 g_6 + i \alpha_6 g_6] e^{i\theta_6}$$

$$A_{g6}^x = A_{C^x} - \omega_6^2 g_6 \cos \theta_6 - \alpha_6 g_6 \sin \theta_6$$

$$A_{g6}^y = A_{C^y} - \omega_6^2 g_6 \sin \theta_6 + \alpha_6 g_6 \cos \theta_6$$

$M_{12} \rightarrow \text{known}$



Link 5



Scanned with CamScanner

Link (6)

Link 2

$$\alpha_2 = 180 - \theta_2$$

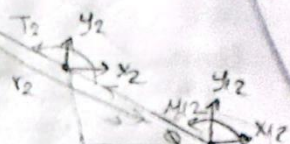
$$x_2 = -m_2 A g_2^x$$

$$y_2 = -m_2 A g_2^y$$

$$T_2 = -I_2 \alpha_2$$

$$x_7 = -m_7 A A^x$$

$$y_7 = -m_7 A A^y$$



$$\sum M_O = 0 = \mu_{12} - x_2 \sin \phi r_{2/2} -$$

$$y_2 \cos \phi r_{2/2} + T_2 - (x_7 - r_{77} \cos \phi) \sin \phi r_2 - (y_7 + r_{77} \sin \phi) \cos \phi r_2 = 0$$

$$F_{47} = 0$$

$$\sum F_y = 0 \rightarrow y_{12} \text{ (4)}$$

$$\sum F_x = 0 \rightarrow x_{12} \text{ (5)}$$

Link 4 $\theta_3 = 180 + \beta$

$$y_4 = -m_4 A g_4^y \quad \beta = \theta_3 - 180 \quad x_4 = -m_4 A g_4^x$$

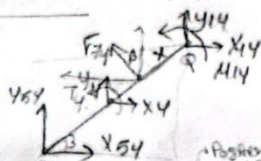
$$F_{74} = -F_{47} \quad T_4 = -I_4 \alpha_4$$

$$\sum M_Q = \mu_{14} + T_4 - y_{54} \cos \beta r_4 + x_{54} \sin \beta r_4 - F_{74} x_4 -$$

$$y_4 \cos \beta r_{4/2} + x_4 \sin \beta r_{4/2} = 0 \rightarrow \text{①}$$

$$\sum F_y = 0 \rightarrow y_{14} + y_4 + y_{54} + F_{74} \cos \beta = 0 \rightarrow y_{14} \text{ (6)}$$

$$\sum F_x = 0 \rightarrow x_{14} + x_4 + x_{54} + (F_{74} \sin \beta) = 0 \rightarrow x_{14} \text{ (7)}$$



Link 5 $\alpha_2 = 360 - \theta_4$

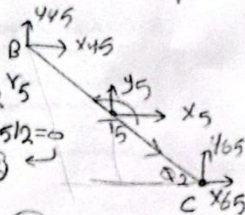
$$\sum M_C = 0 = -y_{45} \cos \alpha_2 r_5 - x_{45} \sin \alpha_2 r_5$$

$$+ T_5 - x_5 \sin \alpha_2 r_{5/2} - y_5 \cos \alpha_2 r_{5/2} = 0$$

$$\text{Solve ① \& ②} \rightarrow y_{45} \text{ \& } x_{45}$$

$$\sum F_y = 0 = y_{45} + y_{65} + y_5 = 0 \rightarrow y_{65} \text{ (8)}$$

$$\sum F_x = 0 = x_{45} + x_5 + x_{65} = 0 \rightarrow x_{65} \text{ (9)}$$

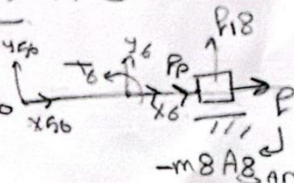


Link 6 $\sum F_y = y_{56} + y_6 + P_{18} = 0$

$$P_{18} \text{ (10)}$$

$$\sum F_x = 0 = -F_P - m_8 A g_8^x + x_6 + x_{56} = 0$$

$$F_P = 0$$



$$\text{I/P Power} = M_{12} \times \omega_2 \text{ watt.}$$

Shaking forces

$$X_{sh} = \sum X_{ii} = -(X_{12} + X_{14} + \delta p)$$

$$y_{sh} = \sum y_{ii} = -(y_{12} + y_{14})$$

$$M_{sh} = -(M_{12} + M_{14})$$

$$\phi_3 = \theta_2 - 90$$

$$F_{OA} = \frac{M_{12}}{\overline{OA}}$$

$$M_{14} = F_{OA} \times \sin \phi_3 \times \overline{OA}$$

